

A.c

```
set ns [new Simulator]
```

```
set tf [open t1a.tr w]
$ns trace-all $tf
```

```
set nf [open t1a.nam w]
$ns namtrace-all $nf
```

```
proc finish {} {
    global ns nf tf
    $ns flush-trace
    execute nam t1a.nam &
    close $tf
    close $nf
    exit 0
}
```

```
set n0 [$ns node]
set n1 [$ns node]
```

```
$ns duplex-link $n0 $n1 3Mb 10ms DropTail
```

```
set udp0 [new Agent/UDP]
$ns attach-agent $n0 $udp0
```

```
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 600
$cbr0 set interval_ 0.005
```

```
$cbr0 attach-agent udp0
```

```
set null0 [new Agent/NULL]
$ns attach-agent $n1 $null0
```

```
$ns connect $udp0 $null0
```

```
$ns at 0.5 "$cbr0 start"
$ns at 4.5 "$cbr0 stop"
$ns at 5.0 "finish"
```

```
$ns run
```

B.c

```
set ns [new Simulator]
```

```
set tf [open t2a.tr w]
```

```
$ns trace-all $tf
```

```
set nf [open t2a.nam w]
```

```
$ns namtrace-all $nf
```

```
proc finish {} {  
    global ns nf tf  
    $ns flush-trace  
    execute nam t2a.nam &  
    close $tf  
    close $nf  
    exit 0  
}
```

```
set n0 [$ns node]
```

```
set n1 [$ns node]
```

```
$ns duplex-link $n0 $n1 3Mb 10ms DropTail
```

```
set tcp0 [new Agent/TCP]
```

```
$ns attach-agent $n0 $tcp0
```

```
set sink [new Agent/TCPSink]
```

```
$ns attach-agent $n1 $sink
```

```
$ns connect $tcp0 $sink
```

```
set ftp [new Application/FTP]
```

```
$ftp attach-agent $tcp0
```

```
$ns at 0.5 "$cbr0 start"
```

```
$ns at 4.5 "$cbr0 stop"
```

```
$ns at 5.0 "finish"
```

```
$ns run
```

C.c

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<unistd.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<arpa/net.h>

int main(int argc,char * argv[])
{
    int cfd;
    char buf[100];
    struct sockaddr_in serv;

    if(argc != 3)
    {
        printf("Usage:%s <IP><PORT>\n",argv[0]);
        return 1;
    }

    cfd = socket(AF_INET,SOCK_STREAM,0);

    serv.sin_family = AF_INET;
    serv.sin_port =htons(atoi(argv[2]));
    serv.sin_addr =inet_addr(argv[1]);

    connect(cfd,(struct sockaddr*)&serv,sizeof(serv));

    printf("Enter Message:");
    scanf("%s",buf);

    write(cfd,buf,sizeof(buf));

    read(cfd,buf,sizeof(buf));

    printf("Server reply:%s\n",buf);
    close(cfd);
    return 0;
}
```

D.c

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
#include<unistd.h>
#include<sys/socket.h>
#include<netinet/in.h>
#include<arpa/net.h>

int main(int argc,char * argv[])
{
    int sfd,cfd;
    char buf[100];
    struct sockaddr_in serv;

    if(argc != 2)
    {
        printf("Usage:%s <PORT>\n",argv[0]);
        return 1;
    }

    sfd = socket(AF_INET,SOCK_STREAM,0);

    serv.sin_family = AF_INET;
    serv.sin_port =htons(atoi(argv[1]));
    serv.sin_addr.s_addr = INADDR_ANY;

    bind(sfd,(struct sockaddr*)&serv,sizeof(serv));
    listen(sfd,5);
    cfd = accept(sfd,NULL,NULL);

    read(cfd,buf,sizeof(buf));
    printf("Client Message:%s\n",buf);
    strcpy(buf,"Message received");
    write(cfd,buf,sizeof(buf));

    close(sfd);
    close(cfd);
    return 0;
}
```

Cisco.c

Building> enable
configure terminal

(config)# interface GigabitEthernet 0/0
(config-if)# ip address 10.11.48.97 255.255.255.240
(config-if)# no shutdown

(config)# interface GigabitEthernet 0/1
(config-if)# ip address 10.11.48.85 255.255.255.224
(config-if)# no shutdown

ASN-3> enable
configure terminal

(config)# interface vlan 1
(config-if)# ip address 10.11.48.114 255.255.255.248
(config-if)# exit

(config)# ip default-gateway 10.11.48.113

(config)# interface vlan 1
(config-if)# no shutdown

IP Address : 10.11.48.62
Subnet Mask : 255.255.255.192
Default Gateway : 10.11.48.1